

Liquidity considerations in estimating implied volatility

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**Indira Gandhi Institute of Development Research, Mumbai
March 2011**

<http://www.igidr.ac.in/pdf/publication/WP-2011-006.pdf>

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Abstract

Some option series in the market are far less liquid than others. Market illiquidity can reduce the informativeness of option prices. In this paper, we propose alternative schemes to estimate implied volatility while reducing the importance attached to illiquid options. Using data for index options traded at the National Stock Exchange in India, we find that the performance of a liquidity weighted scheme is superior to that of more conventional schemes such as the vega weights, the volatility elasticity weights and the traditional vxo. Liquidity weights offers the possibility of improved implied volatility estimation in situations where there is strong cross-sectional variation in option market liquidity.

Keywords:

liquidity, implied volatility, volatility index, Indian index options market

JEL Code:

G12, G13, G17

Acknowledgements:

We are grateful to NSE for the data used in this paper. We thank Nidhi Aggarwal, Ajay Shah, Rajat Tayal and the participants of the IGIDR Finance Research Series for useful discussions. The views expressed in this paper belong to the author and not her employer.

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1 Introduction

Measurement of volatility has always had an important research focus because it is a critical input in financial applications of portfolio allocation and risk management. However, because there was no direct way to observe volatility, proxies of volatility were used instead of a real measure.

When options markets became established and liquid, market price of options were used to directly calculate the market forecast of volatility called the “implied volatility” (IV). More recently, technology improvements in trading and dissemination have led to the availability of price information at very high frequencies. This high frequency data is now used to measure the “realised volatility” (RV) of traded securities from such transparent markets. RV is computed as a sum of squared returns observed intra-day. When sampled sufficiently frequently, it can be arbitrarily close to the underlying integrated volatility which is the integral of instantaneous volatility over a specified period of time.

However, IV has some distinct advantages over the RV of a security, since it is a direct market measure of the *forecast of volatility*. A disadvantage is that there are multiple values for IV calculated from different options on the same underlying and there is no consensus on which gives the best value of IV.

A well documented feature of IV is the *implied volatility smile*, which was the name given to the graph of IV derived from options with varying strikes. The volatility smile has been observed consistently, across different securities that had options traded on them, in different options exchanges across the world. Rubinstein (1994), Jackwerth and Rubinstein (1996), Dumas *et al.* (1998) observe the implied volatility smile for the S&P 500 index options market. Beber (2001) finds a smile profile with negative slope for the Italian market, and Engstrom (2002) reports that the smile profile is characterised by a positive slope for the Swedish market. Pena *et al.* (1999) show that the smile profile is symmetric in the Spanish market.

One endeavour to isolate the true volatility forecast from the multiple values of IV was based on developing models of option pricing that explicitly incorporated the factors that were linked to the IV function, such as moneyness of the option or volatility dynamics or the liquidity underlying options trading. The other endeavour was to develop an index of implied volatility, which is a weighted average of multiple IVs rather than any one.

The focus on an IV index was heightened with the introduction of the CBOE VIX in 1993. This volatility index was calculated for the S&P 100 index

options using the methodology proposed by Whaley (1993). This was disseminated by the CBOE in real time, thus becoming the first exchange to calculate a volatility index (VIX). The index was calculated using only the at-the-money (ATM) options with a defined weighting scheme over the IV values calculated. In response to criticisms, the CBOE shifted the VIX calculation methodology in 2003 to one that calculated VIX by averaging the prices of S&P 500 puts and calls over a wide range of strike prices.

Several exchanges internationally have followed the CBOE in calculating a VIX using market index options prices. However, options market in these economies are highly illiquid when compared to the options market in US. Tzang *et al.* (2010) discuss the illiquidity and unavailability of relevant option prices in constructing a volatility index for an emerging market like Taiwan, and examine the extent to which forecast errors are caused by either the rollover rules or the sampling methods when a volatility index is constructed. It is not well understood whether the new VIX methodology is the optimal one for such markets. In this case, exchanges arrive at different types of approximations to handle problems with missing data.

In this paper, we try to address the problem of illiquid options markets in emerging economies by constructing a weighting scheme that directly incorporates the liquidity of the option. Given access to high-frequency limit order book data on the Indian Nifty (market index) options market, we are able to calculate the spread of each traded option and implement a weighting scheme that adjusts for illiquidity when calculating the VIX. We call this the spread adjusted VIX (SVIX).

We then compare the forecasting performance of SVIX against three weighting schemes used in the literature: the 1993 CBOE index (called VXO), the vega weighted index (VVIX) and a volatility elasticity weighted index (EVIX). The forecastability of each of the VIX candidates is compared with the realised volatility (RV) of the market index every day using the Model Confidence Set (MCS) test (Hansen *et al.*, 2003). We find that the SVIX shows a closer link with the behaviour of the RV than do the other indexes, providing some support for the notion that differences in liquidity across options contribute in part to the volatility smile of the options market. We also run univariate regressions of RV on each volatility index. The regression results indicate that all volatility indexes contain information about future volatility, but that they are biased forecasts.

We present the paper in the following sections: Section 2 presents the issues surrounding the creation of a volatility index. It further discusses the evaluation framework used to compare the performance of our volatility indexes.

Section 3 reviews the literature on different existing methods to construct volatility indexes. It also describes the alternate ways of computing a volatility index. Section 4 contains a description of the data used for the analysis, after which we present the results of our analysis in Section 5. Section 6 concludes.

2 Issues in constructing IV indexes

Gastineau (1977) proposed the use of an index to resolve the problem of multiple values of IVs from different options on the same underlying. Thus, an IV index, which is a weighted average of the IVs from different option prices, would be the summary measure of underlying future volatility.

The first weighting schemes were suggested by Trippi (1977) and Schmalensee and Trippi (1978) which placed equal weights on all the IVs used in calculating the index. However, since the literature showed that the Black-Scholes model apparently priced some options more accurately than others, weighting schemes that had different weights according to different factors were proposed and used. In following years, several researchers made significant progress in developing the concept further (Galai, 1989; Cox and Rubinstein, 1985; Brenner and Galai, 1993; Whaley, 1993).

Commercial interest in an IV index arose when CBOE, in 1993, started disseminating IV index values on the S&P 100 index options. This was called the CBOE VIX. Now the CBOE VIX is synonymous with the measure of market volatility rather than traditional measures such as the volatility of returns of the market index portfolio.

More recently, VIX has garnered a lot of interest because of research showing a negative correlation of the time series behaviour of the VIX values with the market returns. Daigler and Rossi (2006) found significant diversification benefits from adding a long VIX position to an S&P 500 portfolio. Dash and Moran (2005) illustrated potential diversification benefits of adding VIX exposure to hedge fund portfolios. Szado (2009) found that a long exposure to VIX futures and call options could have provided significant portfolio diversification during the 2008 crises. This increased interest in the economic benefits of the VIX has led to a renewed interest in developing volatility indexes for developing markets. This is tempered by two observations about market microstructure conditions in these markets.

The first is the significant increase in the transparency of trades and order

data from the options markets. This feature is similar for both developing and developed markets. This is relevant from the perspective of selecting a set of factors based on which the best set of weights can be designed for the market IV. For instance, the depth of transparency from the electronic limit order book gives the possibility of using direct measures of liquidity for the real-time calculation of the index.

The second is the quality of depth and scope of options trading in developing markets that is far lower when compared with developed markets like the U.S. CBOE. The deployment of the more sophisticated approach that is used in the new CBOE VIX methodology is possible with the level of information available from the developed markets. However, the deployment of the VIX methodology is typically modified to adjust for the smaller options markets in the developing economies.

Thus, the construction of a summary volatility index from options prices in these newer markets may draw upon different weighting schemes depending upon what data is readily accessible. The creation of an optimal methodology then becomes two fold:

1. To identify different weighting schemes based on available data that has a direct influence on the shape of the IV smile.
2. To create a framework using which to identify which is the *optimal* weighting scheme.

2.1 Factors influencing IV values

If the Black-Scholes model held exactly, these options ought to have been priced so that they all have the same implied volatility. However, stylised facts showed this not to be so. The factors identified in the literature across which IV values varied included:

1. Moneyness: how far the strike is from the current spot value.
2. Maturity
3. Sensitivity to volatility, or vega
4. Liquidity

Money/Strike

Some of the first findings about the variation in IV from different options were as a function of strike prices, that came to be known as the *volatility smile*. IV was consistently lower for lower values of the moneyness of the option. Rubinstein (1994), Jackwerth and Rubinstein (1996), Dumas *et al.* (1998) showed that the implied volatility of the S&P 500 index options decreased monotonically as the strike price increased with respect to the current level of the underlying asset.

Maturity

A second consistent feature was the variation with maturity. Prices of near month options showed lower IV than far month options. Heynen *et al.* (1994), Xu and Taylor (1994) and Campa and Chang (1995) show that implied volatilities are a function of time to expiration and thus exhibit a term structure.

Vega

A third consistency arose from the sensitivity of the option to the volatility of the underlying. The derivative of the Black-Scholes price with respect to volatility captures the vega of any given option and is shown to be consistently different for different values of the strike, as well as the maturity of the contract.¹ The vega of an option then naturally lent itself to be used as an input to differentiating the IV of different options when calculating an IV index (Latane and Rendleman, 1976). Chiras and Manaster (1978) suggested weighting by volatility elasticity instead of vega.

Among other influential papers, Beckers (1981) and Whaley (1982) suggested minimizing the weighted sum of the squared deviations between market value and the corresponding Black-Scholes price.²

¹Vega is higher for options that are further away from the money since they have a lower extrinsic value and are less likely to change with changes in implied volatility. It is also higher for options with longer expiration in order to compensate for additional risk taken by the seller.

²This method thus allows the call prices to provide an implicit weighting scheme that yields an estimate of standard deviation which has least prediction error.

Liquidity

Some recent literature has a focus on linking the impact of liquidity on options prices and the IV functions. Brenner *et al.* (2001) show that there is a significant illiquidity premium between two sets of currency options, out of which one set are traded and the other is not. Bollen and Whaley (2004) documented an empirical link between the shape of the IV smile and the depth of the market on the buy and the sell side of options with different moneyness. They show that *net buying pressure* does affect the shape of the IV smile in both the index as well as the single stock options markets. Further, they show that the shape of the IV smile is driven by different market forces for index options compared to single stock options.

Theoretical underpinnings for the link between liquidity and option price start from models of asymmetric information. Nandi (1999) set up a model of asymmetric information linking the level and the shape of IV function to net order flow of options. The model shows that an increase in net options order flow increases the mispricing by the Black-Scholes model. Garleanu *et al.* (2009) formalise the findings in Bollen and Whaley (2004) by incorporating end-user demand in a model for options prices. Here they exploit the feature that end-users tend to hold long index options and short equity options to explain the relative expensiveness of index options. Another model to explicitly incorporate liquidity in the price of stock options was Cetin *et al.* (2006), who show market liquidity premium³ as a significant part of the option price.

Liquidity also explains variation in IV from option with different maturities. Far month options typically have much lower traded volumes than near month options, similar to the liquidity patterns of options with varying degrees of moneyness. An analysis of high frequency order book data from options market shows that far-month options and OTM options have high bid-ask spreads as well.

Etling and Miller (2000) explore the relationship between bid-ask spread as a liquidity proxy with moneyness of options. Even though they find that ATM options have the highest liquidity, they did not find any symmetric variation of liquidity with moneyness. Chou *et al.* (2009) also explore how the IV function varies as a function of liquidity in both the spot and options market. They find that *order based* measures of liquidity (such as the bid-ask spread) better explain the variation in IV than *trade based* measures (such as

³The paper models the liquidity using a generic supply function where option price monotonically increases with size of order.

traded volume). They also find that both spot and options markets liquidity matter for the variation in IV.

Thus, there appear to be multiple factors explaining the shape of the IV function. This, in turn, has led to multiple approaches in designing the weights for an IV index. What is the *efficient* weighting scheme rests upon the performance of the forecast from each scheme against some benchmark volatility measure. We examine the framework to carry out such a performance evaluation of different weighting schemes in the next section.

2.2 Performance evaluation

One of the reasons that there is no consensus on one *best/optimal* weighting scheme for a volatility index is the lack of an observable benchmark volatility. The literature focuses on how to create a framework to evaluate the performance of a volatility forecast in the absence of such a benchmark. For example, such a performance framework is extensively used in testing the forecasts of dynamic volatility models such as GARCH, EWMA, etc. This framework has two broad approaches: one which delivers a relative measure of performance amongst a set of candidate models, and the other which delivers a measure of performance of each of the candidate models against a single benchmark.

Similarly, extensive research has been done to measure the forecast accuracy of IV, i.e., how well IV forecasts the RV of the underlying asset over the life of an option. Most studies use a predictive regression of the IV estimate on future volatility where the goodness of prediction is measured through the coefficients of predictive regressions. Early studies by Day and Lewis (1988), Lamoureux and Lastrapes (1993) and Canina and Figlewski (1993) show that IV is not a good predictor for future return volatilities.

Christensen and Prabhala (1998), Christensen *et al.* (2001) identify and correct some of the data and methodological problems present in early studies. They increase their data span to account for a possible regime shift around the October 1987 crash. They run instrumental variable regressions in addition to OLS regressions to correct for errors-in-variable (EIV) problem in IV. They use non-overlapping samples by using data at a lower frequency (monthly) to avoid the problem of *telescoping overlap* (Christensen *et al.*, 2001) which is about the high degree of overlap in successive *rv* values because the number of observations decrease as the options get closer to expiry.

Latane and Rendleman (1976), Chiras and Manaster (1978), and Beckers

(1981) find that IV performs better in capturing future volatility than standard deviations obtained from historical returns. Blair *et al.* (2001) find that volatility forecasts provided by the early CBOE volatility index are unbiased, and they outperform forecasts augmented with GARCH effects and high-frequency observations. Similar results were reported early on by Jorion (1995) for foreign exchange options.

Corrado and Miller (2005) examine the forecasting quality of three implied volatility indexes based on S&P 100, S&P 500 and Nasdaq 100. They find that the forecasting quality of the volatility index based on the S&P 100 and S&P 500 has improved since 1995, and that those based on the Nasdaq 100 provides better forecasts of future volatility.

The framework of *encompassing regressions* are also used to assess the predictability of IV estimates against other forecast variables. This framework addresses the relative importance of competing volatility forecasts and whether one volatility forecast subsumes all information contained in other volatility forecast(s). Poteshman (2000), Jiang and Tian (2005b), Corrado and Miller (2005) have found that IV estimates are biased, but efficient and informative relative to forecasts from other volatility estimates.

A recent study by Becker *et al.* (2007) used an approach that differs from the traditional forecast encompassing approach used in earlier studies and finds that the S&P 500 IV index does not contain any such incremental information relevant for forecasting volatility. Becker *et al.* (2009) compare the index against a combination of forecasts of S&P 500 volatility by using the Model Confidence Set (*MCS*) methodology and find combination forecasts outperform individual model based forecasts and implied volatility.

In this paper, we use two steps to compare the performance of our volatility indexes:

1. Forecasting regressions following Christensen and Prabhala (1998) to test the information content of the volatility measures. We also run instrumental variable regressions to correct for potential errors-in-variable problems in implied volatility estimates as discussed by previous studies.⁴
2. The Model Confidence Set (MCS) methodology of Hansen *et al.* (2003).

The MCS framework addresses the problem of choosing the *best* forecasting model. It contains the best model with a given level of confidence. It may contain a number of models, which indicates they are of

⁴Christensen and Prabhala (1998), Jiang and Tian (2005b), Corrado and Miller (2005)

equal predictive ability. It has several advantages over other methods like superior predictive ability (SPA) test and the reality check (RC) test.

Hansen *et al.* (2003) discuss three advantages of MCS over tests for SPA. First, that the SPA required a benchmark model to be chosen while the MCS method did not. Second, that the SPA was a measure of relative performance of pairs of models, while the MCS offered evidence of which set of models that outperform other models. Lastly, that the MCS was based on tests of simple hypotheses rather than composite hypotheses which poses problems of interpretation for SPA tests. For example, the RC test of White (2000) uses a composite form of the null hypothesis and is thus sensitive to inclusion of poor (and irrelevant) models in the set of competing models. The test for SPA by Hansen (2001) develops a way to reduce the influence of poor performing models, while the MCS procedure eliminates the poor performing models in the early stage and thus is free from such problems.

The construction of the MCS test is an iterative procedure in that it requires a sequence of tests for equal predictive ability. The set of candidate models is trimmed by deleting models that are found to be inferior. The final surviving set of models in the MCS contain the optimal model with a given level of confidence and are not significantly different in terms of their forecast performance.⁵

The critical question that remains in this is still the choice of the benchmark measure for volatility which we discuss in Section 3.6.

3 Choices of IV indexes

In this section, we describe the different methods we use in order to calculate IV indexes for the Indian options market. More importantly, we also describe the benchmark volatility measure that can be calculated for use in a performance comparison framework so as to identify what might be the *optimal* method to calculate a VIX for India.

We start with a description of the two most widely computed volatility indexes by several exchanges across the world, namely, VXO and VIX.

⁵Hansen *et al.* (2003)

3.1 V XO

This volatility index is calculated using prices of options on the S&P 100 index. The implied volatilities are calculated using the Black-Scholes model, and the VXO is an average of the IVs on eight near-the-money options, including options at the two nearest maturities. When the near-month options reach eight calendar days to expiry, the options used in the calculation are rolled over to the next expiration. At each maturity, the CBOE chooses two ATM calls and puts, where the ATM strikes are determined as those directly above and below the current index level.

VXO takes an average of the two IV estimates from the put and call at each strike price, and then linearly interpolates between the two average IVs at the two strikes prices to obtain the ATM IV. The two ATM IV estimates undergo a trading day conversion, and are further interpolated along the maturity dimension to create a 22-trading-day volatility, which constitutes the VXO.

$$VXO = 100 \times \left[\delta_1 \left(\frac{N_{t2} - 22}{N_{t2} - N_{t1}} \right) + \delta_2 \left(\frac{22 - N_{t1}}{N_{t2} - N_{t1}} \right) \right]$$

where $i = 1, 2$ stands for the *near* and *next* month respectively, δ_i are the related IV, $N_{ti} = N_{ci} - 2\text{int}(N_{ci}/7)$ is the number of trading days to expiration and N_{ci} is the number of calendar days to expiration.

VXO was the first volatility index introduced by CBOE in 1993. In 2003, it was criticised for using an option pricing model and being biased due to the trading day conversion. In addition there were two structural changes⁶ in the US economy that reduced the usefulness of VXO as a measure of future volatility. These were:

1. S&P 500 options became the most actively traded index options.
2. Earlier index calls and puts were equally important in investor trading strategies but in later years the market got dominated by portfolio insurers who bought out-of-the-money and at-the-money index puts for insurance purposes.

Such criticisms of VXO along with changes in the structure of the US options market led to a new approach to calculating the volatility index, VIX, based on the prices of options trading on the S&P 500.

⁶see Whaley (2009)

3.2 VIX

In contrast to the earlier index, (VXO) where IV was derived from options prices using the Black-Scholes options pricing model, VIX has been derived from the concept of fair value of a volatility swap (Demeterfi *et al.*, 1999). Here, even though the variance is derived from market observable option prices and interest rates, the theoretical underpinning is rooted in the broader context of model-free implied variance of Dupire (1993) and Neuberger (1994). This concept was further developed by Carr and Madan (1998), Demeterfi *et al.* (1999) and Britten-Jones and Neuberger (2000). Jiang and Tian (2005a) establish that the variance measure under this framework is theoretically equivalent to the model-free implied variance formulated by Britten-Jones and Neuberger (2000).

The CBOE calculates the model-free implied variance as:

$$\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) \frac{1}{T} \left[\frac{F}{K_0} - 1 \right]^2$$

where $\sigma = \text{VIX}/100$, T = time to expiration, F = forward index level derived from index option prices, K_i = strike price of the i^{th} OTM option, ΔK_i = interval between strike prices half the distance between the strike on either side of K_i , K_0 = the first strike below the forward index level, F , R is the risk-free interest rate to expiration and $Q(K_i)$ is the midpoint of the bid-ask spread for each option with strike K_i .

Calls and puts are included up to the point where there are two consecutive strike prices with a bid price equal to zero. VIX is calculated using the two nearest expiration months of S&P 500 options in order to achieve a 30-calendar-day period. Rollover to the next expiration occurs eight calendar days prior to the expiry of the nearby option. The value of the index is derived from the prices of at-the-money and out-of-the-money calls and puts. The closer is the options strike price to the at-the-money value, the higher is the weight its price receives in the calculation.

Most other markets, including emerging markets, have adopted the new CBOE VIX methodology. However, there are several differences between the emerging markets and the US that suggest the new methodology may not be optimally applied in these markets because of the low liquidity in the options market.

In lieu of the above, we suggest alternate ways of computing a volatility index that overcome the problems of illiquidity and availability of limited strike price in the following sections.

3.3 Volatility linked weights

Following early literature (Latane and Rendleman, 1976; Chiras and Manaster, 1978), we use two different weighting schemes: the vega and the volatility elasticity weighting scheme as choices for an alternative methodology for calculating the market IV index. The schemes are as follows:

1. The Black-Scholes vega to weight the options. The weighting scheme is given as:

$$\sigma_{tj} = \frac{\sum_i w_{it,j} \sigma_{it}}{\sum_i w_{it,j}}$$

where $w_{it,j}$ is Black-Scholes vega for the i^{th} option contract at time t . $j = 1, 2$ denotes the two nearest maturities. And the vega is defined as:

$$\begin{aligned} \text{vega} &= \frac{dC}{d\sigma} \text{ or} \\ \frac{dP}{d\sigma} &= SN'(d_1) \sqrt{(T-t)} \end{aligned}$$

2. There is an alternative to the vega when weighting the option IVs as a function of underlying volatility. The choice is to use volatility elasticities. These are calculated as

$$\sigma_{tj} = \frac{\sum_i w_{it,j} \frac{\sigma_{it,j}}{C_{it,j}} \sigma_{it,j}}{\sum_i w_{it,j} \frac{\sigma_{it,j}}{C_{it,j}}}$$

Here $w_{it,j}$ is Black-Scholes vega and C_{it} is the price for the i^{th} option contract at time t , $j=1, 2$ denotes the two nearest maturities.

Due to the addition of the term $\sigma_{it,j}/C_{it,j}$, the scheme that uses volatility elasticities puts more weight on out-of-the-money options (with low prices $C_{it,j}$) than the vega weights model.

3.4 Liquidity linked weights

The linkages between liquidity and implied volatilities presented in Section 2.1 appear to lead to a calculated VIX value which may be biased due to

illiquidity, low volumes and non-continuous strike prices. Literature has documented that across different underlyings, options on less liquid underlyings have a larger premium compared to those on more liquid underlyings.

We postulate that the same feature of higher premium for illiquid options will also be present for different options on the same underlying. Certainly, illiquidity across different options ends up as a constraint while calculating a volatility index. Jiang and Tian (2005a) found that the VIX constructed by the CBOE is flawed due to truncation errors that arise from the unavailability of option data for very low and very high strikes in practice.

In our work, we adjust for liquidity in two ways. First, we examine the possibility of using order book data (mid-quote price calculated as the midpoint between the bid and ask price in the limit order book (LOB) for the option) rather than traded prices to calculate the option IV. Secondly, we explicitly weight option IV by the available liquidity, which we measure as the bid-ask spread available at that point in time in the limit order book for that option. These weights are calculated as follows:

$$\sigma_{tj} = \frac{\sum_i \frac{1}{w_{it,j}} \sigma_{it,j}}{\sum_i \frac{1}{w_{it,j}}}$$

where $w_{it,j}$ refers to the percentage spread computed for option i at t , and $j = 1, 2$ stands for the two nearest maturities.

The advantage of this method is clearly that liquid options are given more weight, thus reducing biases due to illiquidity. In emerging economies like India, there are limited strikes available to contribute to the calculation of a market volatility index. In applying the methodology of the new CBOE VIX, the lack of availability of options prices traded at a wide range of strikes would magnify the truncation error, and increase the bias of the VIX measure.

In comparison, the method that uses the liquidity weights automatically adjusts for the lack of data by incorporating it in the value of the spread. If there is data missing on either side of the book, the spread would take a value of infinity, and the weight attributable to that option would be zero.

3.5 Adjustment for rollover

The implied volatility estimates obtained for the two nearest maturity is linearly interpolated to obtain a 30 day estimate. Rollover to the next expiration occurs eight calendar days prior to the expiry of the nearby option.

Table 1 Ranked by the number of contracts traded

Rank	Exchanges	Jan-Jun 2009	Jan-Jun 2010	% change
1	Korea Exchange	1,464,666,838	1,781,536,153	21.6%
2	CME Group	1,283,607,627	1,571,345,534	22.4%
3	Eurex	1,405,987,678	1,485,540,933	5.7%
4	NYSE Euronext	847,659,175	1,210,532,100	42.8%
5	National Stock Exchange of India	397,729,690	783,897,711	97.1%

Source: FIA, <http://www.futuresindustry.org/volume-.asp>

The interpolation scheme used is :

$$\text{VIX} = 100 \times \left[\sigma_{t1} \left(\frac{N_{c2} - 30}{N_{c2} - N_{c1}} \right) + \sigma_{t2} \left(\frac{30 - N_{c1}}{N_{c2} - N_{c1}} \right) \right]$$

where σ_{ti} are implied volatilities and N_{ci} is the number of calendar days to expiration. Here, $i = 1, 2$ for the near and next month respectively.

3.6 Benchmark volatility measure

A critical parameter to implement the volatility comparison is the choice of a benchmark volatility measure. The theory of quadratic variation suggests that, under suitable conditions, realised volatility (RV) is an unbiased and highly efficient estimator of volatility of returns (Andersen *et al.*, 2001).

RV is computed as sum of intraday squared returns and is given as follows:

$$\text{RV}_{(t,\Delta)} = \sum_{j=1}^{1/\Delta} r^2(t-1+j\Delta, \Delta)$$

Here $r(t-1+j\Delta, \Delta)$ refers to index returns for small (infinitesimal) time intervals Δ .

Given the availability of data at high frequencies, RV of returns can be readily calculated for any security. We then use these RV measures as our proxy for true volatility and compute the loss under mean square error loss function for each candidate volatility index.

4 Data

We use data on the NSE-50 (Nifty) index options at the National Stock Exchange of India Ltd (NSE). NSE is the fifth largest derivative exchange in

Table 2 Ranked by the number of contracts traded in equity index

Rank	Exchanges	Jan-Jun 2009	Jan-Jun 2010	% change
1	Kospi 200 Options, KRX	1,375,065,894	1,671,466,852	21.6%
2	Emini S&P 500 Futures, CME	308,764,146	299,603,623	3.0%
3	S&P CNX Nifty Options, NSE India	146,706,110	221,430,570	50.9%
4	SPDR S&P 500 ETF Options, CME	181,699,626	219,409,316	20.8%
5	DJ Euro Stoxx 50 Futures, Eurex	178,923,108	205,280,712	14.7 %

Source: FIA, <http://www.futuresindustry.org/volume-.asp>

the world in terms of number of contracts traded. Table 1 shows the number of contracts traded for the period Jan-Jun 2009 and Jan-Jun 2010 for the top 5 derivative exchanges in the world. The number of contracts traded at NSE has almost doubled in Jan-June 2010. It is also the third largest exchange in terms of number of contracts traded in equity index.

Data is available at the level of both intra-day transactions as well as orders. An analysis of a unique market by price (MBP) dataset from NSE (which has data for every order update as well as trade), indicates that on average 65⁷ Nifty options order updates occur per second.

Table 3 shows the average number of records for the Nifty index option contracts from March 2009 to April 2010. The presence of a large number of records on the index options indicates that the options market is active. We compare the graph of IV against moneyness in Figure 1, and percentage spread against moneyness in Figure 2 for both the US and Indian options market. The S&P 500 options are available at 5-point strike intervals and 25-point strike price intervals for far months, while Nifty options are available at 100-point strike price intervals only. As a results, there are a wide range of strikes available for S&P 500 options, while there are only few strikes for Nifty options.

4.1 Data description

In the first stage of the analysis, we focus on computing end-of-day values for the alternative volatility indexes to analyse how they differ in relation to the benchmark volatility measure. Once these have been established, we plan to drill down into high frequency data to further understand what may drive these differences.

⁷An average for the near month option contracts of June 2009. The put average is 52 order updates per second while call average is 75 order updates per second

Table 3 Nifty index options

Month	Avg no. of records
March 2009	2891142
April 2009	3596361
May 2009	2732869
June 2009	3409783
July 2009	4563929
August 2009	4724429
September 2009	4855215
October 2009	4790568
November 2009	6014413
December 2009	7768108
January 2010	7163127
February 2010	8429391
March 2010	7527610
April 2010	8767319
Source: NSE	

The span of data in this analysis covers the period from April 2009 to April 2010. As discussed in Section 3, we aim to compute four volatility indexes. We then plan to compare the dynamic behaviour of these different alternatives with RV as the benchmark measure of market volatility.

For the calculation of the benchmark volatility, we use data on Nifty index price which is available at within-one-second intervals from the trades and orders dataset. As a first step, we discretise this data at 10-minutes. This discretised data is then used to calculate daily market index volatility.

Earlier studies like Canina and Figlewski (1993) use over-lapping samples to evaluate the performance of implied volatility estimates, while other studies like Christensen and Prabhala (1998), Jiang and Tian (2005b), Corrado and Miller (2005) use non-overlapping samples by using data at a lower frequency (monthly) in evaluating the performance of implied volatility estimates. However, since we do not have a longer time series, we restrict ourselves to a daily comparison. For our analysis, all volatility indexes are reduced to daily values by dividing them by the square root of the number of calendar days, 365. Since volatility indexes are ex-ante measures of the volatility, we adjust each day's volatility index to the next period. The adjusted volatility indexes are used further for analysis.

Even though we have both traded prices as well as mid-quote prices available at high frequency, we choose the mid-quote of the bid-ask orders of the options from the options limit order book as the benchmark input price in

the Black-Scholes model to compute IV.

This decision is driven by the illiquidity in the options market. Very frequently, the next-month options market suffers from illiquidity in terms of trades, and there is no traded price that is observable. But there is more liquidity in terms of order updates. Thus, there is information in the order book data that is not reflected in the traded prices. This makes it meaningful to use the mid-quote prices rather than traded prices to calculate IV for the volatility indexes since it reduces the effect of missing data.

The use of mid-quote prices has an automatic liquidity/illiquidity impact in the volatility index calculation – where the options prices are moving due to changes in the limit order book rather than a realised price from a transaction. This problem is prevalent in other emerging markets, and has implicitly driven an incorporation of liquidity considerations into volatility index calculations (Tzang *et al.*, 2010).

In terms of the inputs required for the different volatility index candidates:

- The computation of the alternative indexes requires best buy and sell price, the risk free rate, time to expiration, strike price of all options available for near and second nearby months.
- The traditional VXO requires only ATM options for near-month (one month) and next-month (two-month) maturity.

All the indexes are computed at the close of the market.

There is sufficient empirical evidence which suggests that IV varies with respect to moneyness in many developed and emerging economies, implying different weights for the IV calculated from options with different moneyness. Figure 3 clearly shows that IV in the Indian options market is similarly characterised by a volatility smile.

Another set of weights that have been proposed are based on the vega which is the sensitivity of option prices to changes in volatility as well as liquidity. In Figure 4, we see that ATM puts and calls in the Nifty options markets have a higher vega than ITM and OTM options, which means that these options have a higher sensitivity to volatility.

SVIX is the liquidity weighted volatility index, which has been computed by weighting options IV with liquidity. Here, liquidity is measured as the percentage difference between the buy and sell side prices. Options that are more liquid (i.e., with a low percentage spread) are given more weight in the volatility index computation. Figure 5 shows the relationship between

spread tends to be linear in how it varies with moneyness. From this graph, we infer that a set of weights that are linearly increasing in moneyness is justified. Thus, the IV of each option is adjusted by the inverse of the spread of the option and the weighted average is used to calculate the SVIX.

4.2 Summary of proposed methodology

We propose three alternative weighting schemes through which IV is estimated. The first weighting scheme uses the Black-Scholes vega (VVIX), the second uses volatility elasticity (EVIX) and the third uses percentage spread to weight the options (SVIX) in order to compute a volatility index. The performance of these three volatility indexes, along with the old CBOE methodology (VXO) is compared against the benchmark of realised volatility (RV). We use two steps to compare the performance of our volatility indexes:

1. Run forecasting regressions to test the information content of the volatility indexes. We also run instrumental variable regressions to correct for potential errors-in-variables problems in implied volatility estimates.
2. Use the MCS methodology which allows comparison of multiple volatility forecasts and chooses the volatility forecast which is best in tracking RV.

5 Results and analysis

Table 4 gives the summary statistics for each type of volatility index and log volatility index series. They are all higher on average than the corresponding RV series. The IV values are thus likely biased forecast for RV. In addition, the reported skewness and kurtosis reveal that the log volatility index is more conformable with the normal distribution while the volatility index itself is not.

The graphs in Figure 6 show how each volatility index tracks the RV. All the volatility indexes overestimate the RV. The graph of SVIX vs. RV clearly indicates that it is best in tracking RV followed by VVIX. Differences between RV on day t and IV observed on the day $t - 1$ represent observed forecast errors.

Table 4 Summary statistics of RV and IV

	Volatility					Log Volatility				
	RV	VXO	VVIX	EVIX	SVIX	LRV	LVXO	LVVIX	LEVIX	LSVIX
No. of obs.	216	216	216	216	216	216	216	216	216	216
Min	0.35	1.07	0.88	1.17	0.54	-1.04	0.07	-0.13	0.16	-0.62
Max	3.43	4.00	4.32	3.64	3.41	1.23	1.39	1.46	1.29	1.23
Mean	1.13	2.16	1.78	2.05	1.55	0.02	0.72	0.52	0.68	0.37
Kurtosis	4.85	2.58	4.26	2.68	3.04	2.36	2.24	2.40	2.15	2.65
Skewness	1.14	0.58	1.07	0.74	0.65	0.05	0.03	0.30	0.28	-0.17
Std Dev	0.52	0.66	0.63	0.61	0.56	0.45	0.30	0.34	0.28	0.36
Q1	0.70	1.67	1.31	1.57	1.15	-0.35	0.52	0.27	0.45	0.14
Q2	1.04	1.98	1.57	1.92	1.44	0.04	0.68	0.45	0.65	0.36
Q3	1.45	1.90	2.17	2.35	1.90	0.37	0.97	0.78	0.37	0.64

5.1 Volatility forecast regressions

We run univariate regressions of log RV on each of the log volatility indexes separately to test several hypothesis associated with the information content of the volatility measures. Regressions are run using log volatility series to ensure that the probability density of the error term is close to the normal density.

We consider

$$\text{LRV}_t = a_0 + a_1 \text{LVIX}_{i(t-1)} + \epsilon_t$$

Here LVIX_i belongs to the set of VXO, VVIX, EVIX, SVIX.

If the volatility forecast contains no information about the future volatility then the slope coefficient a_1 would be zero. Table 5 summarises the regression results. The parenthesis below each regression coefficient is the p value. For each regression the t statistic is computed by following a robust procedure taking into account of the heteroscedastic and autocorrelated error structure (Newey and West, 1987). The slope coefficient is positive and significant at 1% for all volatility indexes indicating that all of them contain important information about future volatility. The slope coefficient is not significantly different from one for VVIX, EVIX and SVIX.

If a given volatility forecast is an unbiased estimator of future realised volatility, the slope coefficient a_1 should be one and the intercept a_0 should be zero. The null hypothesis of no bias is tested using the Newey-West covariance matrix. χ^2 statistics and p values are reported in Table 5. The null hypothesis is rejected at the 1% significance level in all cases with estimated coefficients a_1 ranging from 0.95 to 1.17.

Table 5 Regression results

Volatility Indexes	a_0	a_1	Adj. R^2	χ^2	DW
LVXO	-0.83 (0.00)	1.17 (0.00)	0.62	731.1 (0.00)	1.38
LVVIX	-0.50 (0.00)	1.01 (0.00)	0.57	249.1 (0.00)	1.23
LEVIX	-0.69 (0.00)	1.05 (0.00)	0.43	269.0 (0.00)	0.99
LSVIX	-0.33 (0.00)	0.95 (0.00)	0.59	153.5 (0.00)	1.39

Note: p-values in paranthesis

This result is not surprising because summary statistics in Table 4 indicate that all volatility indexes are on average greater than the RV. The evidence is also consistent with the existing option pricing literature which documents that stochastic volatility is priced with a negative market price of risk (or equivalently a positive risk premium). The volatility implied from option prices is thus higher than their counterpart under the objective measure due to investor risk aversion (Jiang and Tian, 2005b).

The adjusted R^2 of VXO is slightly higher than SVIX while VVIX and EVIX have adjusted R^2 lower than SVIX. However, the difference of 3% is not sufficiently high to conclude that VXO has a better predictive ability than SVIX. The DW statistic is significantly different from 2 indicating that the residuals still reflect dependence across time points.

5.2 Instrumental variable regressions

The instrumental variable approach is adopted when there may be possible errors in explanatory variables. Many studies such as Christensen and Prabhala (1998), Jiang and Tian (2005b), Corrado and Miller (2005) have discussed the possible reasons that may result in the error-in-variable problem in implied volatility estimates which may further bias the slope coefficient in the univariate regressions discussed earlier. Following Christensen and Prabhala (1998), we apply a two-stage least squares regression to implement the instrumental variable estimation procedure. We use lagged IV as an instrument for IV. In the first stage, we regress each volatility index on the instrumental variable. In the second stage, RV is regressed on the fitted values obtained from the regression in the first stage.

Table 6 Instrumental variable regression results:

Dependent Variable: $\overbrace{\text{LVIX}_{i(t-1)}}$					Dependent Variable: LRV_t					
1 st Stage	a_0	a_1	Adj.R ²	DW	2 nd Stage	b_0	b_1	Adj.R ²	χ^2	DW
LVXO _{t-2}	0.01 (0.29)	0.98 (0.00)	0.94	2.44	$\overbrace{\text{LVXO}_{t-1}}$	-0.83 (0.00)	1.17 (0.00)	0.59	598.4 (0.00)	1.29
LVVIX _{t-2}	0.04 (0.02)	0.92 (0.00)	0.83	2.68	$\overbrace{\text{LVVIX}_{t-1}}$	-0.52 (0.02)	1.04 (0.00)	0.51	201.8 (0.00)	1.22
LEVIX _{t-2}	0.03 (0.02)	0.95 (0.00)	0.89	2.63	$\overbrace{\text{LEVIX}_{t-1}}$	-0.73 (0.00)	1.10 (0.00)	0.43	243.5 (0.00)	0.97
LSVIX _{t-2}	0.02 (0.02)	0.94 (0.00)	0.87	2.65	$\overbrace{\text{LSVIX}_{t-1}}$	-0.34 (0.00)	0.97 (0.00)	0.54	147.3 (0.00)	1.29

We consider the following:

$$\overbrace{\text{LVIX}_{i(t-1)}} = a_0 + a_1 \text{LVIX}_{i(t-2)}$$

$$\text{LRV}_t = b_0 + b_1 \overbrace{\text{LVIX}_{i(t-1)}} + \epsilon_t$$

Table 6 summarises the results for the instrumental variable regressions. We find no material change in statistical inferences between instrumental variable and OLS regressions.

5.3 MCS results

Table 7 reports the rankings of all log volatility indexes based on mean square error (MSE), as well as the MCS results. The MSE errors are the first column of data in the table which shows that LSVIX has the smallest MSE and is thus the most accurate forecast of future volatility.

The remainder of the columns show the MCS results. Here, it turns out that the model with the largest range statistic T_r is VXO. The p-value in the first reduction is 0.019. As it is eliminated in the first round, this automatically determines that the MCS p-value for VXO is 0.019.

In the second round EVIX is eliminated with a p-value of 0.011. Since this p-value is smaller than the MCS p-value of model previously dropped hence the MCS p-value for EVIX is 0.019.

Table 7 Loss function rankings and MCS results

VIX	MSE	p_{T_r}	$\text{MCS}(p_{T_r})$	$p_{T_{SQ}}$	$\text{MCS}(p_{T_{SQ}})$
LVXO	0.392	0.019	0.019	0.000	0.000
LEVIX	0.304	0.011	0.019	0.000	0.000
LVVIX	0.201	0.006	0.019	0.006	0.006
LSVIX	0.112	-	1.000	-	1.000

In the third round VVIX is eliminated with a p-value of 0.006. As this p-value is smaller than the MCS p-value of model(s) previously dropped the MCS p-value for VVIX is 0.019. The results remain the same when we look at the semi-quadratic statistic T_{SQ} rather than T_r .

The only volatility index that survives in the MCS is SVIX while all other volatility indexes are dropped at 5 % level of significance.

6 Conclusion

The VXO and VIX are widely accepted volatility indexes and are computed by many exchanges across the world. While VIX may be a good indicator of investor sentiment in a developed economy like US, it may not be an optimal estimate in an emerging economy like India where there is insufficient liquidity across the options contracts traded on the market index, Nifty, to support the calculation of the Indian market measure of volatility.

The CBOE methodology uses near and next month option contracts to compute the implied volatility index. Low volumes for second nearby Nifty option contracts and availability of limited strike price might induce a bias in its computation for an emerging market like India.

To overcome such problems, our study suggests alternate ways of computing implied volatility indexes and compares their performance in capturing future volatility. These are based on simpler methodologies that incorporate the wide variation in liquidity of Nifty options, such as the vega-weighted and the elasticity-weighted index. In addition, we propose a volatility index that is weighted by measures of liquidity from the limit order books of the contracts traded in the options market on Nifty.

Once we have calculated these different measures of implied volatility indexes, we test their performance using two procedures: forecasting regressions and the MCS methodology. In each case, we use the realised volatility of the

underlying using one-minute Nifty returns observed using intra-day Nifty levels.

The forecasting regressions indicate that all volatility indexes contain important information about future volatility. However, they are biased forecasts. We then use the MCS methodology that allows comparison of multiple forecasts to select the volatility index that is best at tracking the realised volatility for Nifty. The results indicate that the liquidity weighted volatility index (SVIX) outperforms other volatility indexes in capturing future volatility of Nifty.

References

- Andersen TG, Bollerslev T, Deibold FX, Labys P (2001). “The Distribution of Realized Exchange Rate Volatility.” *Journal of the American Statistical Association*, **96**, 42–55.
- Beber A (2001). “Determinants of the implied volatility function on the Italian Stock Market.” *LEM Papers Series 2001/05*, Laboratory of Economics and Management (LEM), Sant’Anna School of Advanced Studies, Pisa, Italy.
- Becker R, Clements AE, White SI (2007). “Does implied volatility provide any information beyond that capyured in model-based volatility forecasts?” *Journal of Banking and Finance*, **31**, 2535–2549.
- Becker R, Clements AE, White SI (2009). “Are combination forecasts of S&P 500 volatility statistically superior?” *International Journal of Forecasting*, **24**, 122–133.
- Beckers S (1981). “Standard deviations implied in option prices as predictors of future stock price variability.” *Journal of Banking and Finance*, **5**, 363–381.
- Blair BJ, Poon SH, Taylor SJ (2001). “Forecasting S&P 100 volatility: The incremental information content of implied volatilities and high-frequency index returns.” *Journal of Econometrics*, **105**, 5–26.
- Bollen NPB, Whaley RE (2004). “Does Net Buying Pressure Affect the Shape of the Implied Volatility Functions?” *Journal of Finance*, **LIX**(2), 711–753.
- Brenner M, Eldor R, Hauser S (2001). “The price of options illiquidity.” *Journal of Finance*, **LVI**(2), 789–804.
- Brenner M, Galai D (1993). “Hedging Volatility in Foreign Currencies.” *The Journal of Derivatives*, **1**, 53–59.
- Britten-Jones M, Neuberger A (2000). “Option prices, implied price processes, and stochastic volatility.” *Journal of Finance*, **50**, 507–528.
- Campa JM, Chang PHK (1995). “Testing the Expectations Hypothesis on the Term Structure of Volatilities in Foreign Exchange Options.” *Journal of Finance*, **50**, 529–47.

- Canina L, Figlewski S (1993). “The information content of implied volatility.” *The Review of Financial Studies*, **6**, 659–681.
- Carr P, Madan D (1998). “Towards a theory of volatility trading.” *Risk*.
- Cetin U, Jarrow R, Protter P, Warachka M (2006). “Pricing options in an extended Black-Scholes economy with illiquidity: Theory and empirical evidence.” *The Review of Financial Studies*, **19**(2), 492–529.
- Chiras DP, Manaster S (1978). “The information content of option prices and a test of market efficiency.” *Journal of Financial Economics*, **6**, 213–234.
- Chou RK, Lin Chung S, Hsiao YJ, Wang YH (2009). “The impact of liquidity risk on option prices.” *Technical report*, National Central University, Taiwan and National Taiwan University, Taiwan.
- Christensen BJ, Hansen CS, Prabhala NR (2001). “The Telescoping Overlap Problem in Options Data.” *Working Paper*, University of Aarhus and University of Maryland.
- Christensen BJ, Prabhala NR (1998). “The relation between realized and implied volatility.” *Journal of Financial Economics*, **50**, 125–150.
- Corrado CJ, Miller TW (2005). “The forecast quality of CBOE implied volatility indexes.” *Journal of Futures Markets*, **25**, 339–373.
- Cox JC, Rubinstein M (1985). *Options Markets*. Prentice-Hall, Englewood Cliffs, N.J.
- Daigler RT, Rossi L (2006). “A Portfolio of Stocks and Volatility.” *Journal of Investing*, **15**, 99–106.
- Dash S, Moran MT (2005). “VIX as a Companion for Hedge Fund Portfolios.” *The Journal of Alternative Investments*, **8**, 75–80.
- Day TE, Lewis CM (1988). “The behavior of the volatility implicit in the prices of stock index options.” *Journal of Financial Economics*, **22**, 103–122.
- Demeterfi K, Derman E, Kamal M, Zou J (1999). “A guide to volatility and variance swaps.” *Journal of Derivatives*, **6**, 9–32.
- Dumas B, Fleming J, Whaley RE (1998). “Implied Volatility Functions: Empirical Tests.” *Journal of Finance*, **53**, 2059–2106.

- Dupire B (1993). “Model art.” *Risk*, pp. 118–120.
- Engstrom M (2002). “Do Swedes smile? On implied volatility functions.” *Journal of Multinational Financial Management*, **12**, 285–304.
- Etling C, Miller TW (2000). “The relationship between index option mon-eyness and relative liquidity.” *The Journal of Futures Markets*, **20**(10), 971–987.
- Galai D (1989). “A Proposal for Indexes for Traded Call Options.” *The Journal of Finance*, **34**, 1157–1172.
- Garleanu N, Pedersen LH, Poteshman AM (2009). “Pricing options in an extended Black-Scholes economy with illiquidity: Theory and empirical evidence.” *The Review of Financial Studies*, **22**(10), 4259–4299.
- Gastineau GL (1977). “An Index of Listed Option Premiums.” *Journal of Quantitative Finance and Accounting*, **33**, 70–75.
- Hansen PR (2001). “A test for superior predictive ability.” *Brown University, Department of Economics Working Paper*.
- Hansen PR, Lunde A, Nason JM (2003). “Choosing the best volatility mod-els: The model confidence set approach.” *Oxford Bulletin of Economics and Statistics*, **65**, 839–861.
- Heynen R, Kemna A, Vorst T (1994). “Analysis of the Term Structure of Implied Volatilities.” *Journal of Financial and Quantitative Analysis*, **29**, 31–56.
- Jackwerth JC, Rubinstein M (1996). “Recovering probability distributions from option prices.” *The Journal of Finance*, **51**, 1611–1631.
- Jiang GJ, Tian YS (2005a). “Gauging the investor fear gauge: Implementa-tion problems in the CBOE’s new volatility index and a simple solution.” *Technical report*.
- Jiang GJ, Tian YS (2005b). “The Model-Free Implied Volatility and Its Information Content.” *Review of Financial Studies*, **18**, 1305–1342.
- Jorion P (1995). “Predicting volatility in the foreign exchange market.” *Journal of Finance*, **50**, 507–528.
- Lamoureux CG, Lastrapes WD (1993). “Forecasting stock-return variance: Understanding of stochastic implied volatilities.” *The Review of Financial Studies*, **6**, 293–326.

- Latane HA, Rendleman RJ (1976). “Standard Deviations of Stock Price Ratios Implied in Option Prices.” *The Journal of Finance*, **31**, 369–381.
- Nandi S (1999). “Asymmetric information about volatility: How does it affect implied volatility, options prices and market liquidity?” *Review of Derivatives Research*, **3**(3), 215–236.
- Neuberger A (1994). “The log contract: A new instrument to hedge volatility.” *Journal of Portfolio Management*, **2**, 74–80.
- Newey WK, West KD (1987). “Simple Positive Definite Heteroskedasticity and Autocorrelation Consistent Covariance Matrix.” *Econometrica*, **55**, 703–708.
- Pena I, Rubio G, Serna G (1999). “Why do we smile? On the determinants of the implied volatility function.” *Journal of Banking and Finance*, **23**, 1151–1179.
- Poteshman AM (2000). “Forecasting future volatility from option prices.” *University of Illinois at Urbana-Champaign, Working Paper*.
- Rubinstein M (1994). “Implied binomial trees.” *The Journal of Finance*, **49**, 771–818.
- Schmalensee R, Trippi R (1978). “Common Stock Volatility Expectations Implied by Option Premia.” *The Journal of Finance*, **33**, 129–47.
- Szabo E (2009). “VIX Futures and options: A Case Study of Portfolio Diversification During the 2008 Financial Crisis.” *The Journal of Alternative Investments*, **12**, 68–85.
- Trippi RR (1977). “A Test of Option Market Efficiency Using a Random-Walk Evaluation Model.” *Journal of Economics and Business*, **29**, 93–98.
- Tzang SW, Hung CH, Wang CW, Shy DSD (2010). “Do liquidity and sampling methods matter in constructing volatility indices? Empirical evidence from Taiwan.” *International Review of Economics and Finance*.
- Whaley RE (1982). “Valuation of American call options on dividend-paying stocks : Empirical tests.” *Journal of Financial Economics*, **10**, 29–58.
- Whaley RE (1993). “Derivatives on Market Volatility: Hedging Tools Long Overdue.” *The Journal of Derivatives*, **1**, 71–84.

- Whaley RE (2009). “Understanding VIX.” *The Journal of Portfolio Management*, **35**(3), 98–105.
- White H (2000). “A reality check for data snooping.” *Econometrica*, **68**, 1097–1126.
- Xu X, Taylor SJ (1994). “The Term Structure of Volatility Implied by Foreign Exchange Options.” *Journal of Financial and Quantitative Analysis*, **29**, 57–74.

Figure 1 Comparing the variation of implied volatility by moneyness: An example

These graphs show the relationship between implied volatility and moneyness for the US vs. the Indian market index options markets as observed on 11/03/2011. The first is the plot of the *put options market* on the S&P 500 at the Chicago Board Options Exchange (CBOE) and the Nifty at the National Stock exchange (NSE), where the data depicted is for all puts that have an expiration date of 15th April 2011 for the US market, and 28th April 2011 for the Indian market. On the y-axis is the implied volatility (%) and on the x-axis is the moneyness of the option, calculated as $(\text{Strike} - \text{Current index level})/(\text{Strike})$ and also expressed in %. Similarly, the second is the plot of the *call options market* for the market index at CBOE and the NSE where moneyness of the option is calculated as $(\text{Current index level} - \text{Strike})/(\text{Strike})$ and expressed in %. The graphs show that the US markets have a far greater set of strikes at which options trade compared to the Indian markets. This is one reason that an automatic adoption of the CBOE VIX may not be the optimal methodology to calculate the market VIX for India as an emerging market.

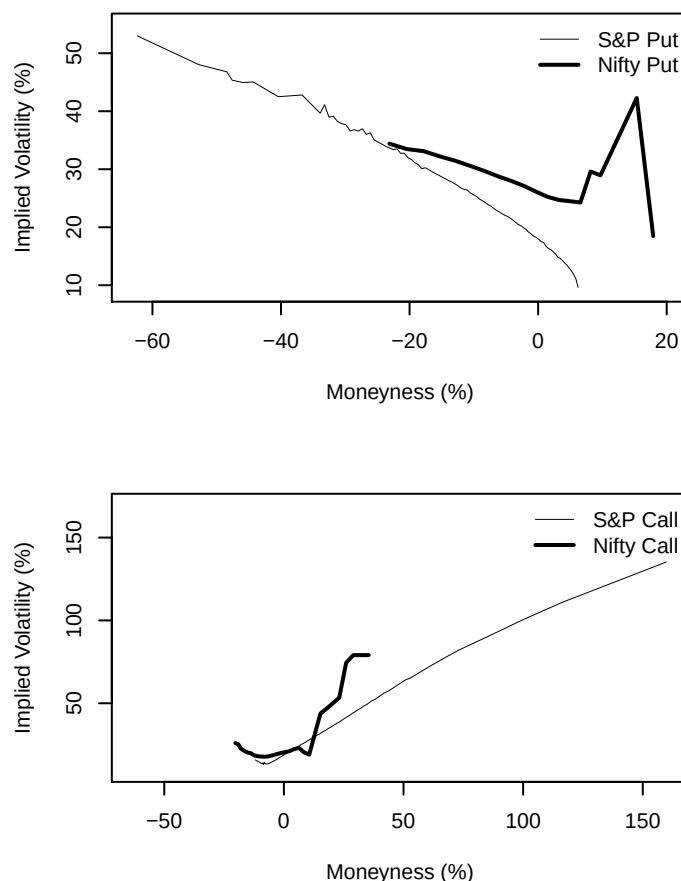


Figure 2 Comparing the variation of liquidity by moneyness: An example

These graphs show the relationship between percentage spread and moneyness for the US vs. the Indian market index options markets as observed on 11/03/2011. The first is the plot of the *put options market* on the S&P 500 at the Chicago Board Options Exchange (CBOE) and the Nifty at the National Stock exchange (NSE), where the data depicted is for all puts that have an expiration date of 15th April 2011 for the US market, and 28th April 2011 for the Indian market. On the y-axis is the percentage spread and on the x-axis is the moneyness of the option, calculated as $(\text{Strike} - \text{Current index level})/(\text{Strike})$ and expressed in %. Similarly, the second is the plot of the *call options market* for the market index at CBOE and the NSE where moneyness of the option is calculated as $(\text{Current index level} - \text{Strike})/(\text{Strike})$ and expressed in %.

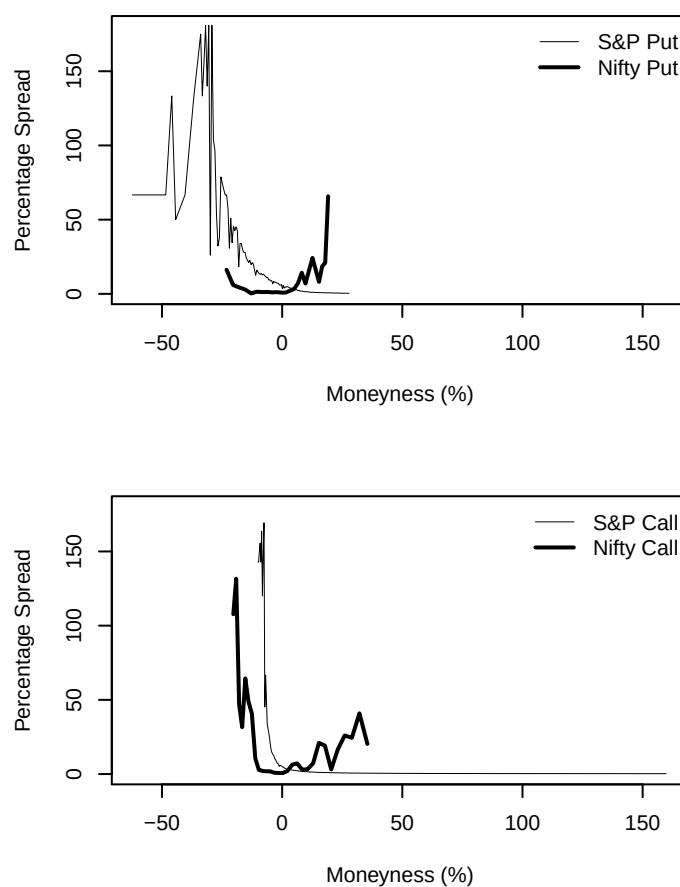


Figure 3 Implied volatility vs. moneyness of Nifty options, February 2010

The graph is computed from options data for the month of February 2010 for near month contracts.

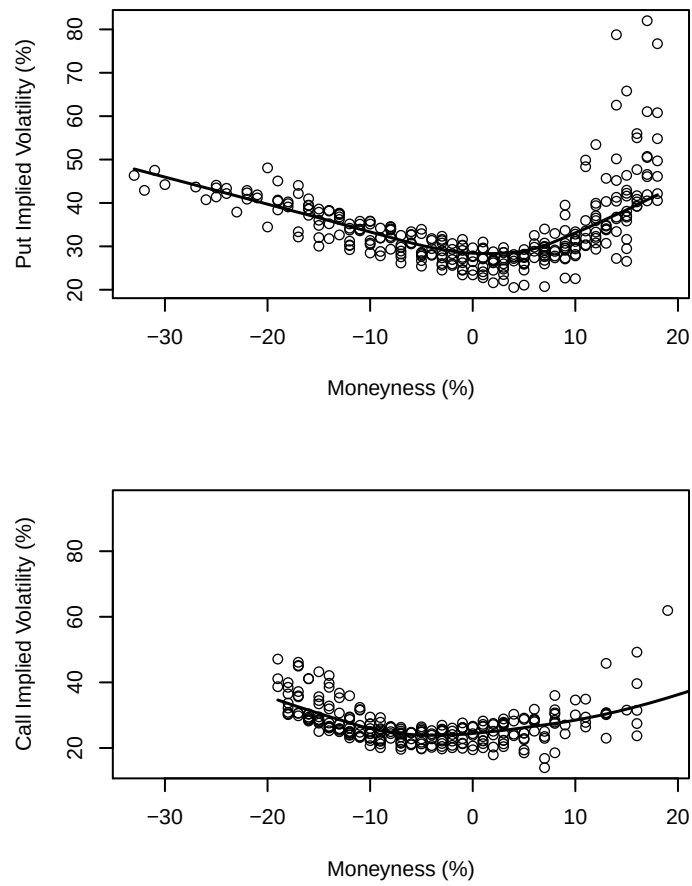


Figure 4 Vega vs. moneyness of Nifty options, February 2010

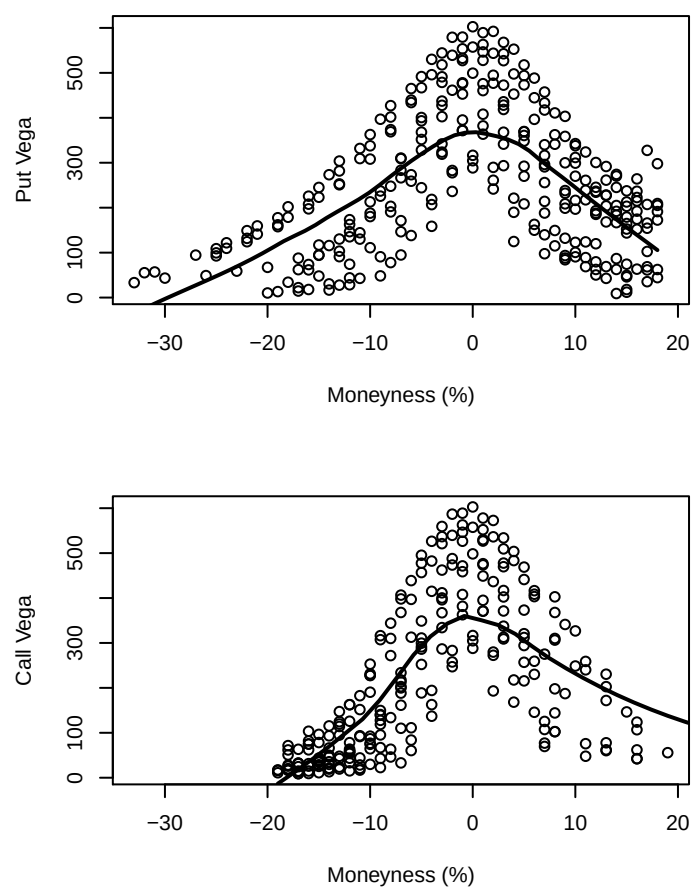


Figure 5 Percentage spread vs. moneyness of Nifty options, February 2010

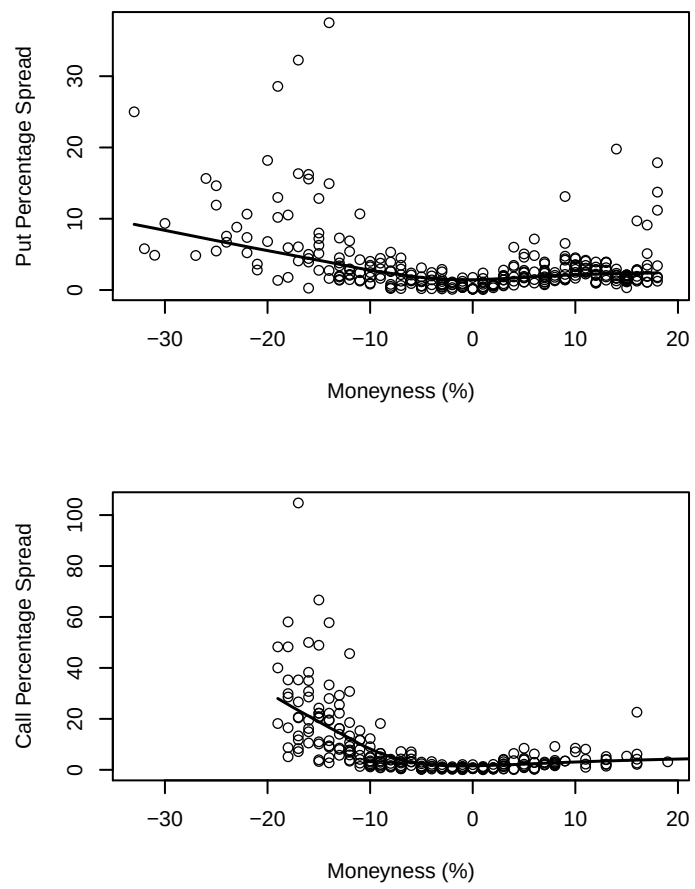


Figure 6 Volatility indexes vs. RV

